



Free education policy and rural Residents' happiness: Financial perspective

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ABSTRACT

Using data from the 2010–2015 China General Social Survey, we applied a propensity score matching difference-in-differences model to examine how a 12-year free-education policy reform affected the happiness of Chinese rural residents. We found that the 12-year free-education pilot will make rural residents happier if the local government's spending on education does not reduce spending on other public services. We also found that the policy has benefited the beneficiaries' grandparents, high-income rural families, and relatively well-educated groups mainly by improving family finances and health. We recommend expanding the pilot program to increase the number of beneficiaries. We suggest that China would be in a position to fully implement the 12-year compulsory education only when the central and local governments can support its implementation.

1. Introduction

Government-provided free-education services play a significant role in improving family happiness (Jin & Liu, 2023). Research suggests that food and clothing, compulsory education, basic medical care, and housing security are considered basic needs of families. As the economy and society develop, the quantity and quality of residents' basic needs will increase. In turn, as a government's financial resources increase, its ability to meet residents' basic needs and social welfare will increase (Frijters et al., 2020). Maslow's hierarchy of needs suggests that when a family's basic needs are consistently met, their overall happiness increases (Drakopoulos & Karayiannis, 2007). Therefore, by providing free education, the government helps better fulfill a family's educational needs without adding financial strain, thereby increasing the families' happiness (French & Vigne, 2019).

The theory of educational justice provides a key theoretical basis for governments to offer compulsory education. The duration and quality of compulsory education and the equality between urban and rural areas can significantly affect residents' sense of educational attainment and happiness. A question then arises: How should the government ensure compulsory education? According to the theory of educational justice,

the sufficiency view suggests that the government must ensure that all children receive at least a minimum (compulsory) education (Tooley, 1995). The egalitarian view emphasizes equal opportunities for rural and urban children to access a minimum standard of education (John Rawls, 2017). The preferentialist view suggests that resources should be directed toward promoting disadvantaged children's wellbeing (Brighthouse, 2002; Schouten, 2012).

The government's financial resources constrain the extension of the period of compulsory education in China. Education serves as the foundation of science and technology, which are key drivers of economic and social development. In China, education is considered crucial for the country's rejuvenation. The Chinese Communist Party and government considers nine years of compulsory education as essential to achieving its strategic goal of "rejuvenating the country through science and education." By the end of 2011, China had achieved universal nine-year compulsory education. Since 2006, pilot programs for free senior high school education have been launched in selected counties and cities in Shaanxi, Guangdong, Fujian, Hebei, and the Inner Mongolia provinces. Since 2008, there have been repeated proposals to legislate the extension of the nine-year compulsory education period. Implementing a 12-year free-education system could effectively address the current low

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senior high school enrollment rates in rural and impoverished areas of central and western China. This initiative would also optimize the distribution of educational resources and ultimately promote equal access to educational opportunities. However, given China's current economic challenges, extending the period of compulsory education may further strain the existing fiscal revenue–expenditure imbalance.¹ The 12-year compulsory education reform cannot be immediately implemented; the level of subsidies and the scope of the pilot should be gradually broadened commensurate with the growth of financial resources.

Assessing the impact of the 12-year free-education policy reform on the wellbeing of rural residents and conducting a path analysis can provide crucial insights for decision making on compulsory education policy reform. Wellbeing is a key objective in the formulation, implementation, and evaluation of public policies (Frijters et al., 2020). The Chinese Communist Party and government prioritize maximizing everyone's happiness. As 33.84 % of China's population still lives in rural areas as of 2023 and their happiness levels are reportedly low, improving their wellbeing and overall societal happiness has become a key focus and challenge for the Party and the government. Education significantly influences residents' happiness, yet there is a noticeable rural–urban disparity in educational resources in China. According to egalitarian education theory, the government must ensure that all rural children receive the same educational opportunities as those in urban areas (John Rawls, 2017). It remains unclear whether education truly improves the happiness of rural residents in practice. Existing literature mainly explores how individual education levels influence happiness (Araki, 2022; Michalos, 2008; Nikolaev, 2016; Yu, 2014). Some studies focus on the impact of public education spending on happiness (Yin et al., 2019; Zhu & Yu, 2023), while others examine how educational policy reforms affect the wellbeing of rural residents. However, these studies mainly examine the effect of higher education expansion on rural residents' happiness (Jiang, 2022; Nikolaev, 2016; Sanfo et al., 2024). Limited research has specifically addressed the impact of extending compulsory education on rural residents' happiness (Dursun & Cesur, 2016). This study uses data from the 2010–2015 China General Social Survey (CGSS) to evaluate changes in wellbeing among rural residents before and after the implementation of the 12-year free-education policy in China's pilot areas. The study empirically examines how education affects rural residents' happiness and offers valuable insights for decision making on the timing of implementing the reforms.

This study makes several key contributions. First, this study is the first to examine the effect of the 12-year free-education policy on residents' happiness. By analyzing changes in rural residents' happiness before and after the policy's implementation in pilot areas and analyzing whether government can financially support the free-education reform, we discuss the optimal timing for implementing the 12-year compulsory education reform.

Second, current discussions on whether and when to implement the 12-year compulsory education policy mainly focus on its financial feasibility, the effectiveness of the nine-year compulsory education policy, and the duration of compulsory education in other countries and regions. However, there is a lack of empirical data supporting the 12-year compulsory education. This study provides empirical evidence on the impact of extending compulsory education from the perspective of residents' happiness. It highlights that extending the compulsory

education relieves the financial burden on families and improves their education level, which can improve their job market competitiveness, increase income, and improve health.

The remainder of the study is presented as follows. Section 2 reviews the related literature and outlines the research hypotheses. Section 3 describes the data and methodology. Section 4 presents the empirical analysis, and Section 5 offers a summary and conclusions.

2. Literature and hypotheses

2.1. Literature review

Education directly impacts residents' happiness, with this direct impact seen on both psychological and social levels. According to positive psychology, happiness stems from people's pursuit of spiritual life and yearning for the spiritual world. The essence of education is to transform people's subjective experiences and promote spiritual satisfaction, thereby contributing to increased happiness (Hu & Gao, 2019). In social work, individuals with higher education levels often gain more freedom from menial work, leading to a sense of self-control and happiness (Easterlin, 2001). However, empirical studies on the relationship between education level and residents' wellbeing have not aligned with this theoretical analysis. Education can reduce individuals' sense of happiness (Witter et al., 1984) because highly educated people often have greater expectations of themselves, which can lead to increased tension, stress, and anxiety in work and life (Clark & Oswald, 1994). Further, people with higher levels of education have a stronger sense of superiority. When faced with unexpected situations, such as temporary unemployment, they experience a greater sense of relative deprivation than others, thus affecting their perception of happiness (Layard, 2006). Previous research has suggested that a U-shaped relationship between education and happiness, with happiness initially increasing with education and then decreasing (Hartog & Oosterbeek, 1998; Yu, 2014). As higher education becomes more widespread and skills become more common, the marginal effect of university education on happiness gradually diminishes (Araki, 2022). The turning point in this relationship often occurs around university education (Michalos, 2004; Qiu & Zhang, 2021) or secondary education (Stutzer, 2004).

Conflicting perspectives on education and happiness are often due to different research subjects between different countries and groups (Akaeda, 2019). Among the rural population of modern China, there are often empty nesters and left-behind children who remain in rural areas, as most young and middle-aged people work as migrant workers in cities to support their families. Previous studies on the impact of education on the happiness of rural elderly residents have found that the educated rural elderly residents experience happiness levels that are 158 % higher than those of uneducated counterparts (Xiang & Li, 2020). Moreover, higher education levels are associated with greater happiness among unmarried rural elderly people (Zhou, 2021). Besides, children's higher education makes the rural elderly residents happier (Xu & Huang, 2015). Research on how education affects migrant workers' happiness shows that those with secondary education (e.g., senior high school or junior college) are happier (Sun et al., 2022; Wu, 2007). The new generation of migrant workers with a junior high school education is more likely to become self-employed, those with a high school education tend to become employers, and those with a college or bachelor's degree or above tend to become employees (Zhao & Zhou, 2019). Less educated migrant workers (e.g., junior high school level) often struggle to find good job opportunities. Conversely, those with higher education (e.g., college level and above) have higher expectations for vocational positions and tend to be less happy if these expectations are unmet. Existing studies have focused on how education influences the happiness of rural

¹ The average subsidy standard for free education in regular high schools in Tangshan is 1000 yuan per year. Inner Mongolia offers 2000 yuan per year and provides free textbooks. The subsidy standard in Zhuhai is 1620 yuan per year, while the range in Ningbo city varies from 1400 to 2300 yuan per year. Based on an estimated average of 1500 yuan per student annually, multiplied by the total of 41.63 million high school students, China's financial investment needs to increase by over 60 billion yuan each year to implement the 12-year compulsory education policy effectively. It is obvious that the reform cannot be achieved overnight.

elderly and migrant workers. However, the effects of 12 years of free education extend beyond the direct beneficiaries to their family members' (i.e., the indirect beneficiaries) feelings of wellbeing and even to potential beneficiaries of the policy.² From the happiness perspective, these findings are crucial for a comprehensive evaluation of the policy's effectiveness and serve as key for deciding whether implementing 12 years of compulsory education is necessary. Existing studies on the highest level of happiness among migrant workers with secondary education levels and whether the findings are universal among all rural residents provide empirical evidence for deciding from a happiness perspective the need to extend compulsory education from 9 to 12 years.

Education has an indirect effect on rural residents' happiness through several factors and channels. Human capital theory suggests that workers can enhance their human capital through more education, thereby improving their job competitiveness (Schultz, 1990). Increased job competitiveness directly results in higher income, better welfare benefits, lower work intensity, and lesser unemployment risk. For rural residents, the main goal is to secure a stable income, which is the main reason for migrant workers to seek employment in cities (Research Group of the Development Research Center of the State Council, 2011). Education can therefore increase rural residents's happiness by improving employment quality and income levels. Increased income can be invested or consumed (e.g., the purchase of commercial housing) (Yu & Cai, 2019), which enhances consumers' sense of happiness by increasing their utility. Human capital theory also suggests that education and health can partially substituted for each other (Rosenzweig & Schultz, 1989) and that people with higher education have greater cognitive and adaptive abilities, more health awareness, and can better optimize their health investment portfolio, thereby improving their health (Grossman & Kaestner, 1997): the healthier the body, the greater the level of happiness (Bukenya et al., 2003). Moreover, mental health has a greater impact on happiness than physical health (Li & Wang, 2015). People with higher education have a more independent attitude toward marriage and feel less social pressure (Zhao & Liu, 2019), which helps them find a spouse who aligns more closely with their ideals (Gesthuizen et al., 2008), increasing their likelihood of happiness. Thus, education indirectly enhances rural residents' happiness through employment, income, housing, health, and marriage. Empirical studies have shown that education can increase income through employment, which in turn makes the rural residents in China happier. However, only the eastern region in China has confirmed the indirect impact of education on rural residents' happiness through improved health conditions and satisfying housing preferences (Huang et al., 2017).

The effects of education on migrant workers' happiness are less indirect, with more than 90 % of the impact being direct. This contrasts sharply from the direct and indirect effects of education on the settled urban workers' happiness levels (Zhou, 2015) mainly because of the migrant workers' low overall education level. Education mainly helps migrant workers improve their sense of self-identity and ability to

recognize new things. However, its role in securing ideal jobs and increasing income is relatively small. Therefore, the long-term investment value of education is low, and education tends to directly improve rural residents' happiness.

Some studies (Huang et al., 2017) have shown that education indirectly reduces migrant workers' happiness. Higher education makes migrant workers compete with urban workers, yet face difficulties in career promotion because of the urban-rural labor divide. Higher education level increases the migrant workers' sense of integration into the city, leading them to seek more support for children's education services and social security policies, along with lower living costs; therefore, a simple residence no longer meet their expectations for urban integration. As education level increases, the number of migrant workers reduce over time. However, migrant workers who move to cities do so mainly to make money and place less emphasis on leisure, which can lead to less happiness. Understanding how education affects the rural residents' happiness and exploring its impact on different income groups can help accurately implement the 12-year free-education policy and enhance its effectiveness.

2.2. Hypothesis development

According to Maslow's hierarchy of needs theory, once a family's basic food and clothing needs are met, ensuring access to education, basic medical care, and housing security becomes increasingly urgent. A government satisfying these needs directly increases family satisfaction. Additionally, according to the sufficient educational justice theory, it is government's responsibility to ensure that all children receive a minimum appropriate level of education (Tooley, 1995). To date, China has eradicated poverty. Most rural residents in China traditionally aspire for their sons' success and power and daughters' greatness. The transition from nine-year compulsory education to twelve-year free education can reduce rural family's education financial burden, increase the likelihood of rural children pursuing higher education, and contribute to the rural families' wellbeing. Accordingly, we propose the following hypothesis:

Hypothesis 1. The implementation of a 12-year free-education policy in China is expected to significantly impact the happiness of rural residents.

Different age groups of rural residents may experience varying degrees of happiness due to this policy. Parents, as guardians, often bear the burden of their children's high school tuition and miscellaneous fees and a reduced burden or exemption should positively impact parental happiness. However, most young and middle-aged people from rural areas work outside their hometowns throughout the year and only provide regular financial support for living expenses, including their children's education costs. Consequently, the high school tuition subsidy policy might not significantly impact working parents' happiness as it does not substantially reduce their economic pressures. In contrast, older adults who remain in rural areas to take care of their grandchildren are directly responsible for household expenditures and benefit greatly from the high school tuition subsidies their grandchildren receive. These subsidies through 12 years of free education directly reduce families' education economic burdens and subsequently enhance their overall happiness. Therefore, we propose the following hypothesis:

Hypothesis 2. The 12-year free-education policy has a significant impact on the grandparents' happiness, who are beneficiaries, to other age groups, while parents also experience notable variations in their happiness levels.

Rural residents with varying family economic conditions have different perceptions of happiness toward the policy. Low-income rural groups, who occupy a lower socioeconomic position and face challenging living conditions, have a strong desire to improve their children's prospects through education. They are willing to provide them better educational opportunities to enhance their quality of life.

² The implementation of the 12-year free-education policy will benefit three types of groups: the first is the direct beneficiaries of the policy—quasi-high school students and high school students. The second is the indirect beneficiary of the policy—the caregivers of the beneficiary students. Because the 12-year free-education policy is implemented through economic subsidies that reduce tuition and miscellaneous fees, this will directly reduce the family education expenses of policy beneficiaries. Therefore, caregivers of policy beneficiaries can benefit indirectly from the policy implementation. The third is the potential beneficiaries of the policy—other relevant personnel. The 12-year free-education policy is highly notable in promoting social equity, enhancing the quality of citizens, and enhancing comprehensive national strength. The resulting policy externalities will benefit others, and hence, other members of society will be potential beneficiaries of the policy. As the data obtained are not longitudinal tracking data, the research objects of this study are the indirect beneficiaries and potential beneficiaries of the policy, rather than the children who directly enjoy 12 years of free education.

However, due to their income level, they are unable to afford high school tuition and miscellaneous fees beyond the 9-year compulsory education period. After implementing the 12-year free-education policy, low-income families receive a direct educational subsidy, which eases their education financial pressure and reduces their sense of relative deprivation from being at the lower end of the social income structure. Consequently, low-income families tend to be happier with the implementation. For high-income groups, the policy implementation can help in saving part of their education expenses, which increases their disposable income and, in turn, helps improve their quality of life and happiness. However, because the proportion of children's education expenditure in rural high-income family's expenditure is slightly lesser than those in low-income groups, the happiness resulting from saving in education expenses for rural high-income groups is not as significant as that of low-income groups. Accordingly, we propose the following hypothesis:

Hypothesis 3. Compared with the rural high-income group, the 12-year free-education policy has a more obvious positive effect on the happiness of the rural low-income group.

Rural residents with varying education levels perceive the impact of the 12-year free-education policy on their happiness differently. As noted earlier, there is a U-shaped relationship between education and happiness, indicating that this impact may diminish beyond a certain threshold. Less educated rural residents may have limited benefits from limited education and may not adequately comprehend the implications of 12-year compulsory education because of their low education and cognition, which may lead to distrust about the 12-year compulsory education policy. Conversely, those rural residents who have benefited from 9 years of compulsory education may believe that 12 years of compulsory education can make them happier. With higher education, rural residents acquire greater cognitive abilities and dialectical thinking skills, helping them better understand the other potential consequences arising from this policy implementation, such as whether it has crowded out other public services for farmers. Accordingly, this study proposes the following:

Hypothesis 4. The higher the rural residents' education level, the more significant the positive happiness effect of the 12-year free-education policy; however, beyond a certain threshold of educational attainment, the happiness effect of the policy declines.

The 12-year free-education policy affects rural residents' happiness in several ways.³ As noted earlier, education increases happiness within certain limits (e.g., a high school degree or above). This study focuses on rural residents. Currently, rural residents' average years of education have not reached the inflection point of the inverted U-shaped curve⁴; therefore, extending the duration of education improves their happiness. This study shows that education affects rural residents' happiness in the following ways. First, education helps increase rural residents' income by enhancing their employment competitiveness. As income increases, their housing conditions and quality of life improve, which improves their happiness. Second, higher education encourages more compatible

partners, thereby elevating the experience of happiness. Finally, better education can improve individuals' health awareness, which improves their health status and happiness. Accordingly, we hypothesize the following:

Hypothesis 5. Enhanced education can increase rural residents' happiness by improving income, marital status, and health.

3. Empirical strategy, data, and variable construction

3.1. Empirical strategy

This study examines the effect of implementing a 12-year free-education policy on rural residents. Owing to the differences in the policy pilots and implementation times of various prefecture-level cities and the characteristics of multiperiod cross-sectional data, following existing research (Courtemanche & Zapata, 2014), we construct a propensity score matching difference-in-differences (PSM-DID) model as follows:

$$H_{ijt} = \alpha_0 + \alpha_1 Policy_{ijt} + \theta Macro_{it} + \lambda Micro_{ijt} + f_t + \zeta_{it} + \varepsilon_{ijt} \quad (1)$$

The dependent variable H_{ijt} is the happiness level of rural residents in the j th sample of city i in a period t . The core independent variable $Policy_{ijt}$ is the interaction term of the policy variable and the time variable, which is used to measure whether the 12-year free-education policy was implemented in region i in a period t . α_1 is the core independent variable coefficient, and $\alpha_1 > 0$ implies that the policy makes rural residents happier. In contrast, $\alpha_1 < 0$ implies decreased happiness levels of rural residents. $Macro_{it}$ is the macro control variables of city i in year t , including GDP per capita, fiscal expenditure per capita, and the urbanization rate. $Micro_{ijt}$ is microcontrol variables that reflect individual characteristics, including gender, ethnicity, age, religion, employment status, income status, health status, housing status, marital status, and educational satisfaction. f_t and ζ_{it} represent the fixed effects of year and region, respectively, and ε_{ijt} is the random perturbation term.

3.2. Data and sample

We use two databases to construct an empirical sample. We first use the CGSS to draw our sample, which included data from 2010 to 2013 and 2015 because the 2014 data and the prefecture-level coding information of the data after 2015 were unavailable. The samples were collected from the multilayered hierarchical cross-sectional survey data of 51,574 urban and rural residents over the age of 18 in 32 provinces (municipalities and autonomous regions) in China. We collected 23,050 valid samples of rural residents.⁵ We use the *China Urban Statistical Yearbook* as the second database to collect data on macro control variables from these yearbooks.

3.3. Variable selection and descriptive statistics

Happiness of rural residents in prefecture-level cities before and after

³ The limited implementation period and the absence of longitudinal data preclude a direct examination of whether the 12-year free-education policy impacts respondents' happiness through changes in their income, health, and marital status. Therefore, this study examines the impact of the educational attainment of previous generations of rural residents on their current income, health, marital status, and other socioeconomic conditions using large-scale cross-sectional data. Consequently, it establishes a general framework illustrating the causal pathway through which years of education influence the wellbeing of rural inhabitants.

⁴ According to the China Human Capital Report 2021, the average years of education of the national labor force increased from 6.1 years in 1985 to 10.5 years in 2019, with urban areas experiencing a rise from 8.2 years to 11.4 years and rural areas seeing an increase from 5.5 years to 9.1 years.

⁵ We clean the data as follows: First, because the research object is rural residents, the sample of urban household registration is excluded. Second, samples missing key variables such as subjective wellbeing, education level, and resident income were excluded. Then, because the CGSS database randomly selects family members over 18 years old from each family as survey objects, students under 18 years old were not included in the sample range, so samples with children were withdrawn. Since our research object is not a child who is a direct beneficiary, the sample included the oldest (95 years old). Finally, a few samples of extreme values were eliminated. Overall, 23,050 effective samples of rural residents were obtained, and 17,346 effective samples satisfying the common trend hypothesis were matched by the propensity score matching method.

the policy reform is the dependent variable. These data are drawn from the CGSS questionnaire and measured based on the following question: “Overall, how do you feel about the life you live?” The answers “very unhappy,” “unhappy,” “general,” “happy,” and “very happy” are assigned values of 1, 2, 3, 4, and 5, respectively.

The 12-year free-education policy is the core independent variable. By the end of 2015, 18 regions (including county-level and prefecture-level cities) in the sample cities had implemented the 12-year free-education policy at different times.⁶ Considering that the CGSS no longer publishes county-level data, only the prefecture- and city-level pilot cities are retained as experimental groups. We take the pilot areas that can obtain happiness data as the experimental group, which are assigned a value of 1, and the other areas are included in the control group, which are assigned a value of 0. Table 1 shows the final characteristics of the participants in the experimental group.

Table 2 shows the summary statistics of the variables used in this study. (1) The final sample included more women than men. The male-to-female ratio was 48:52. (2) Most People aged 40–60 years were classified as middle-aged or elderly. The age range of the respondents was between 18 and 95 years, with both the mean and median being approximately 49 years. The highest proportion was 51.59 % for middle-aged and elderly people (age range 40–60 years), 24.77 % for those aged ≥60 years, and only 26.24 % for those aged 18–40 years. This characteristic agrees with the current rural population in China, as most middle-aged and people under 40 years have left their hometowns to

Table 1
Cities and samples in the experimental group.

Province	Specific cities	Pilot time	Observations	Percentage (%)
Fujian Province	Mawei district and Fuzhou city	2010	250	17.35
	Zhangzhou development zone	2014	46	3.19
Guangdong Province	Shenzhen	2013	9	0.62
Hebei Province	Tangshan city	2008	243	16.86
Shaanxi Province	Yan'an city	Before 2010	473	32.82
		Before 2010	212	14.71
Zhejiang Province	Yinzhou district and Ningbo	Before 2010	212	14.71
Inner Mongolia Autonomous Region	Total autonomous region	2012	208	14.43
Total			1441	100

⁶ The pilot areas are Yinzhou District of Ningbo City, Wuyi County of Shaanxi Province, Zhuhai City of Guangdong Province, Qian'an City of Hebei Province, Tangshan City of Hebei Province, Xizang Autonomous Region, Dexing City of Jiangxi Province, Shuangliu County of Chengdu City, Xinzheng County of Henan Province, Jishou City of Hunan Province, Shaoshan City of Hunan Province, Jinjiang City of Fujian Province, Mawei District of Fuzhou City, Inner Mongolia Autonomous Region, Nanjiang three prefectures (Hotan, Kashi, and Kezi Prefecture), Zhangzhou City of Fujian Province, Nanjing City in Jiangsu Province, and Shenzhen City in Guangdong Province. The regions excluded owing to the lack of CGSS data are Qian'an City of Hebei Province, Dexing City of Jiangxi Province, Jishou City of Hunan Province, Shaoshan City of Hunan Province, Zhuhai City of Guangdong Province, Jinjiang City of Fujian Province, Xinjiang, and Xizang. However, Nanjing has also implemented a pilot program for 12 years of free education, extending preschool education through the distribution of early childhood education vouchers, which is quite different from other places. Therefore, it is not included in the experimental group.

work. (3) The duration of education is low, with 6.64 years as the average years of education in the sample. Among respondents born after 1970, 50.86 % had received nine years of compulsory education⁷ and 18.81 % had a high school degree or above. Overall, there was a low penetration rate of high school education. (4) Most participants were married, with a mean of 1.94, showing that most of the survey sample had a family or a partner. (5) The proportion of farmers engaged in agriculture was the highest, and the proportion of migrant workers was relatively low. According to the CGSS questionnaire “Work Experience and Status,” 44.59 %, 26.77 %, and 28.64 % of the respondents whose work status was “current farming,” “current no job,” and “current nonfarming,” respectively. This indicates a fewer nonagricultural migrant workers in the sample, implying the reality that migrant workers had left the area seeking work and rarely appeared in the survey. Table 2 shows the variables and descriptive statistics.

4. Empirical results

4.1. Preliminary test of the impact on the happiness of rural residents

We first examined how the 12-year free-education policy affects the overall happiness of rural residents in pilot areas. We applied the PSM method to obtain comparable samples between the experimental and control groups. Year-by-year PSM was used to identify similar individuals as those in the pilot area each year. These matched individuals served as “counterfactual” estimates for those in the experimental group, with corresponding variables used as covariates for the PSM. We calculated the propensity score using a binary dependent variable model logit regression and adopted the 1-to-2 nearest-neighbor matching method within a 0.05 caliper range to match similar samples in all samples. After matching, 17,346 valid samples satisfied the common trend hypothesis. As each pilot city's respondents were correlated to the policy evaluation, we regarded the disturbance terms of the same city as a whole and performed city-level clustering processing on the data. Simultaneously, we controlled for the year- and region-fixed effects to reduce the missing variable deviation. To improve the reliability of the experimental results, we used the PSM-DID, supplementary DID, and ordered probit DID methods in the benchmark regression.

Table 3, column 1, presents the OLS common DID regression results for all pilot areas. Column 2 shows the ordinal probability DID regression results for all pilot areas, and columns 3–5 show that for all pilot areas after annual PSM matching. Column 3 shows the samples with nonempty weights, column 4 presents the samples that satisfy the common support hypothesis, and column 5 shows the samples that use frequency-weighted regression.

Table 3 shows that for the full sample, the implementation of the 12-year free-education policy negatively impacts the overall happiness among rural residents in pilot areas, which supports Hypothesis 1. However, the direction of the impact contradicts our expectations. We attribute this outcome to several factors. First, not all rural residents in pilot areas benefited from the policy due to age restrictions, which can lead to feelings of unfair exclusion from subsidies. For example, families whose children began working before high school miss out on the annual public education subsidies, which amounts to thousands of yuan, potentially leaving them deprived. Additionally, people who did not anticipate the policy reforms did not have the foresight enroll their children in high school recently may be upset on not receiving education subsidies. Furthermore, even within families receiving free high school education subsidies, family members' perception about the benefits may differ. This problem is closely examined in the following heterogeneity analysis. Second, spending on 12 years of free education may have

⁷ Further statistics are conducted on people born after 1970, because they were of middle school age and were the first group to receive policy benefits when the 9-year free-education policy was implemented in 1986 in China.

Table 2

Variable definitions and descriptive statistics.

Variables	Index interpretations	Observations	Average	Standard	Minimum	Maximum
Happiness	Very unhappy = 1, unhappy = 2, unhappily unhappy = 3, relatively happy = 4, very happy = 5	23,050	3.783	0.858	1.000	5.000
Policy	The 12-year free-education policy “implementation group” = 1 and “control group” = 0	23,050	0.063	0.242	0.000	1.000
Gender	Male = 1 and female = 2	23,050	1.523	0.499	1.000	2.000
Yearold	The year of the CGSS minus the respondent’s year of birth	23,050	49.115	13.815	18.000	95.000
Nation	Han = 1 and ethnic minority = 0	23,050	0.898	0.303	0.000	1.000
Religion	No religious affiliation = 1 and religious affiliation = 0	23,050	0.866	0.341	0.000	1.000
EdU	Years of schooling	23,044	6.634	3.854	0.000	21.000
Employ	1–6 point rating variables	23,048	2.529	1.389	1.000	6.000
Per-income	Household per capita income/10,000	20,773	1.166	3.491	0.000	400.000
Health	The health level corresponds to an evaluation score between 1 and 5; the higher the score, the healthier the person	23,040	3.446	1.165	1.000	5.000
House	Number of house properties	20,922	1.109	0.827	0.000	96.000
Marriage	Unmarried = 0; divorced, separated, widowed = 1; and married, cohabited = 2	23,050	1.940	0.253	0.000	2.000
EdU-satisfy	Corresponding to the evaluation score between 1 and 5; the higher the score, the more satisfactory	17,818	3.675	1.233	1.000	5.000
Per-GDP	GDP/total population at the end of the year	22,489	4.077	2.622	0.719	46.774
Urban	Urban population/total population at the end of the year	23,050	50.829	9.237	22.67	89.6
Per-finance	Government expenditures/total population at the end of the year	23,050	0.779	0.304	0.363	2.643

Table 3

Baseline regression and annual PSM-DID results.

	(1)	(2)	(3)	(4)	(5)
	OLS	Oprobit	Weight! = .	On_Support	Weight_Reg
Policy	−0.2625*** (−5.7386)	−0.3941*** (−4.5174)	−0.3673*** (−3.7912)	−0.4154*** (−4.1384)	−0.4071*** (−3.7583)
Gender	0.0812*** (5.6454)	0.1128*** (5.7356)	0.1891*** (2.9756)	0.1064*** (4.9583)	0.2050*** (3.2150)
Yearold	0.0088*** (9.4685)	0.0122*** (10.0405)	0.0128*** (6.0043)	0.0129*** (10.1439)	0.0161*** (5.7471)
Nation	−0.0066 (−0.1909)	−0.0059 (−0.1303)	0.0409 (0.2516)	0.0145 (0.3158)	0.0715 (0.4592)
Religion	−0.0417* (−1.6690)	−0.0699** (−2.0549)	−0.1770** (−2.5189)	−0.0800** (−2.1662)	−0.1804** (−2.2485)
EdU	0.0160*** (5.8628)	0.0196*** (5.3215)	0.0251*** (3.1721)	0.0180*** (4.4883)	0.0283*** (3.0765)
Employ	0.0190*** (2.9051)	0.0252*** (2.8058)	−0.0187 (−0.7009)	0.0181* (1.7878)	−0.0138 (−0.4868)
Per-income	0.0055 (1.1522)	0.0068 (1.0622)	0.0489*** (3.5321)	0.0275*** (3.1609)	0.0563*** (3.6810)
Health	0.1947*** (20.2314)	0.2599*** (20.0423)	0.2817*** (9.1762)	0.2647*** (19.1520)	0.2628*** (8.3449)
House	0.0984*** (7.7604)	0.1364*** (8.0558)	0.1868*** (4.2642)	0.1494*** (7.5632)	0.1787*** (3.6790)
Marriage	0.2340*** (6.9018)	0.2859*** (6.6820)	0.1671** (2.1831)	0.3043*** (6.8914)	0.2015*** (2.5853)
EdU-satisfy	0.0825*** (7.9915)	0.1184*** (8.6473)	0.0964*** (2.6276)	0.1264*** (8.4988)	0.0952*** (2.6635)
Per-GDP	−0.0340* (−1.9724)	−0.0522** (−2.0921)	−0.0247 (−0.7490)	−0.0485* (−1.7198)	−0.0417 (−1.3117)
Urban	−0.0209 (−1.5364)	−0.0381* (−1.9407)	−0.0168 (−0.4055)	−0.0347* (−1.9017)	−0.0403 (−0.9278)
Per-finance	−0.2178 (−0.4500)	−0.5370 (−0.8097)	−0.7461 (−0.4242)	−0.6731 (−0.8016)	−0.2803 (−0.1495)
Cons	2.6519*** (3.6939)				
Year fixed effect	Yes	Yes	Yes	Yes	Yes
Regional fixed effects	Yes	Yes	Yes	Yes	Yes
Urban clustering	Yes	Yes	Yes	Yes	Yes
Observations	20,508	20,508	2945	17,346	3601
Adj. R ²	0.1284	0.0620	0.0764	0.0658	0.0760

In this table, we treat the disturbance term of the same city as a whole and cluster the data at the city level considering the correlation of policy evaluations between respondents in each pilot city. T-statistics are reported in parentheses. *, **, and *** indicates significance at the 10 %, 5 %, and 1 % level, respectively.

squeezed out spending on other public services, resulting in its positive effect being offset by the negative effect of reduced public services, thereby resulting in a negative overall net effect. This can be verified indirectly through regional sample regression. In regions with relatively

high education expenditure⁸ and an overall growth rate of fiscal expenditure is not lower than that of education expenditure (such as Tangshan and Zhangzhou) (Fig. 1), there is a significantly positive happiness effect of education policy reform (Table 4). Overall, the timing to fully implement the 12-year compulsory education policy may not be right. However, once the number of policy beneficiaries increases with the expansion of the pilot policy and when both the central and local financial resources are sufficient to support the 12-year free-education policy, it will be the perfect time to fully implement the 12-year free-education policy. In addition, the baseline regression results show that other factors (including education level, employment status, health status, housing status, and marital status) are positively significant for rural residents' subjective wellbeing, consistent with the existing literature.

4.2. Heterogeneity analysis

To further determine which groups experience greater happiness from the 12-year free-education policy, we conducted a heterogeneity analysis among different age, income, and education level groups.

4.2.1. Happiness effects across age groups

Table 5 presents the happiness effects of the 12-year free-education policy among rural residents of different ages. In columns 1–7, the happiness effects of people aged 39–42 and 57–60 are 0.213 and 0.378, respectively, while those of other age groups are negative. According to China's Fifth National Population Census in 2000, rural women bore their first child at 24.02 years.⁹ According to the policy's direct beneficiaries, junior high school (quasi-high school) and high school students are 13–18 years. The people aged 39–42 years are equivalent to their parents, and the people aged 57–60 years are equivalent to their grandparents. This evidence supports Hypothesis 2. Implementing the 12-year free-education policy has increased the happiness of the beneficiary students' parents and grandparents, and that of their grandparents is greater than that of their parents.

4.2.2. Happiness effects across income groups

Table 6 shows the effect of the 12-year free-education policy on the happiness of rural residents with varying income levels. In columns 1–3, the coefficients of the policy do not exhibit "propensity" characteristics. The higher the income of rural residents, the happier they are with the policy implementation; however, the happiness among low-income groups has decreased significantly due to the policy implementation. Therefore, Hypothesis 3 is not supported. The low-income groups are less happy because although their children receive subsidies for part of their tuition and fees, they still must bear accommodation and living expenses. To ease financial pressure, they may prefer their children to start working as soon as possible to support their families. These families are particularly unhappy because they are aware of the policy but cannot benefit from it. Columns 4 and 5 provide empirical evidence from the policy coefficients when household income is reclassified into quintiles. For the very low-income class at the 0%–20% level, even with a free-education policy, children of poor families may not continue to high school. Therefore, despite the policy, their happiness level may be low.

⁸ The average proportion of education expenditure in seven cities from 2010 to 2015 was 20.93 % in Fuzhou, 19.27 % in Tangshan, 18.55 % in Zhangzhou, 17.85 % in Yan'an, 15.69 % in Ningbo, 13.8 % in Inner Mongolia, and 13.15 % in Shenzhen.

⁹ The data in this study are sourced from CGSS from 2010 to 2015. Since most of the students who benefit from the policy during the survey period are children born in the late 1990s and early twenty-first century, it is appropriate to use the first childbearing age of their mothers, ~2000.

4.2.3. Happiness effects across education level groups

Table 7 shows the effect of the 12-year free-education policy on the happiness of rural residents with different education levels. In columns 1–5, the policy coefficients show that only the group with a high school education has a positive happiness effect. After policy implementation, the happiness of the high school education group increased by 0.7528 points, on average, while that of the uneducated group decreased by 0.3952 points.¹⁰ The happiness levels of the groups with junior middle school or primary school education decreased by 0.5866 and 0.9006 points, respectively. The happiness effect on the groups with college-level or higher education was negative but not statistically significant overall, thereby confirming Hypothesis 4. The happiness effect of the 12-year free-education policy reform shows an inverted U-shaped relationship with educational level among rural residents. From an economic perspective, high school graduates have higher-income potential than those with only junior high school education or less, owing to their higher knowledge and skills leading to better employment opportunities and, therefore, an improved sense of happiness. From a societal perspective, individuals with high school education may experience higher social status and recognition within rural communities. This increase in social capital boosts their sense of self-worth and psychological satisfaction. As education levels increase, so do expectations for improved career prospects and higher incomes. However, in a competitive job market, unmet expectations can reduce the positive impact of free education. For example, the enrollment scale of China's universities has increased sharply from 1.08 million in 1998 to 10.42 million in 2023. The expansion has made university education more accessible, thereby diluting the happiness returns associated with it. Consequently, while the overall effect of this policy on happiness is negative for people with college-level and higher education, it is not statistically significant. Overall, rural residents' educational level influences their assessment of educational policy initiatives. Education can enhance the rural residents' psychological and social wellbeing. In particular, the 12-year free-education policy has a substantially improves the happiness of people with high school educations. Therefore, from a happiness perspective, it is essential to extend compulsory education from 9 years to 12 years.

4.3. Mechanism analysis

We used the stepwise regression method to test the mediating role of employment, income,¹¹ health, and marriage in the effect of education on residents' subjective happiness. Table 8 shows that, given the same individual characteristics and urban characteristics, for every year of additional education, the employment score increases by 0.083 points, personal income increases by approximately 3.65 %, the health score increases by 0.0139 points, and the probability of finding a spouse and marrying increases by 0.27 %.

Table 9 shows the impact of education on happiness through employment, income, health, and marriage. We conducted the ordered probit regression without considering employment, income, health, or marriage (column 1). However, when we added personal income into the model in column 2, the regression coefficient of education degree decreased from 0.0268 in the benchmark regression to 0.0203, while

¹⁰ The happiness of the uneducated group does not decrease the most owing to policy implementation, possibly because a notable percentage of this group are grandparents of students who benefit from the policy. In the sample, 33.8 % of the grandparents (57–60 years old) who improve their happiness considerably do not receive education. They have raised the overall happiness level of the uneducated group, relatively reducing the negative impact of policy implementation on the overall happiness of this group.

¹¹ To study the transmission mechanism from education to happiness based on individual level, we replace the variable "per capita annual income of families" in the previous section with the variable "annual income of individuals."

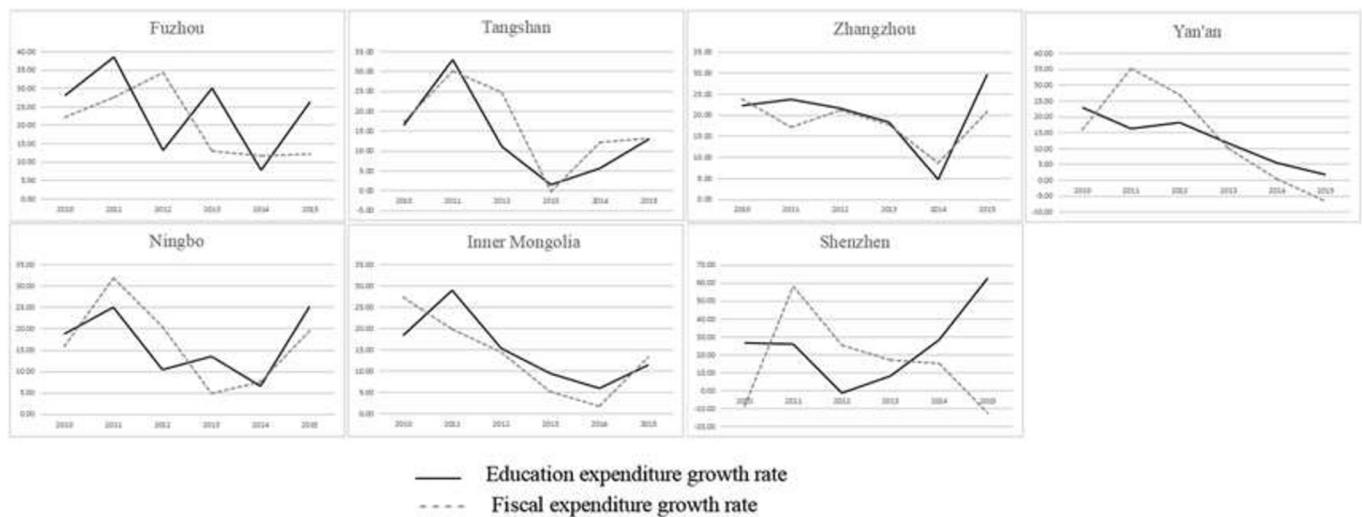


Fig. 1. Comparison of the growth rates of both education and fiscal expenditures in pilot areas. Note: Source from the *China Statistical Yearbook*.

Table 4

Annual PSM-DID regression results for different region groups.

	(1) Fuzhou	(2) Tangshan	(3) Zhangzhou	(4) Yan'an	(5) Ningbo	(6) Inner Mongolia	(7) Shenzhen
Policy	−0.3336*** (−2.6384)	0.1169** (2.4980)	0.3780*** (3.5790)	−0.0917** (−2.5304)	−0.5941*** (−6.9787)	−0.5126*** (−11.7136)	–
Other variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Regional fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Urban clustering	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	17,346	17,346	17,346	17,346	17,346	17,346	17,346
Adj. R ²	0.0524	0.0522	0.0522	0.0522	0.0529	0.0525	0.0522

T-statistics are reported in parentheses. *, **, and *** indicate significance at the 10 %, 5 %, and 1 % level, respectively.

Table 5

Annual PSM-DID regression results by different age groups.

	(1) Age 18–29 years	(2) Age 30–38 years	(3) Age 39–42 years	(4) Age 43–49 years	(5) Age 50–56 years	(6) Age 57–60 years	(7) Age ≥ 61 years
Policy	−1.2510*** (−7.0582)	−1.0211* (−1.9179)	0.2134 (0.7291)	−0.6958*** (−5.3073)	−0.1149 (−0.5106)	0.3785** (2.5385)	−0.6580*** (−4.2064)
Control variables	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Regional fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Urban clustering	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1182	2754	1920	3470	2573	1562	3885
Adj. R ²	0.1387	0.0880	0.1070	0.0904	0.0932	0.0983	0.0825

T-statistics are reported in parentheses. *, **, and *** indicate significance at the 10 %, 5 %, and 1 % level, respectively.

that of personal income was 0.1653 and significant at the 1 % level. This confirms that the influence of education on rural residents' happiness is partly caused by personal income. When we added employment status based on the education–income–happiness channel to column 3, the regression coefficient of degree of education decreased from 0.0202 in Column 2 to 0.0018, that of personal income decreased from 0.1653 in Column 2 to 0.1622, and that of employment status was 0.0235 and significant at the 1 % level. This shows that employment status partly affects the influence of income on rural residents' happiness. When we added health status to the benchmark model in column 4, the regression coefficient of education degree decreased from 0.0268 to 0.0216, while that of health status was 0.2624 and significant at the 1 % level. This proves that health status partly causes the influence of education on

rural residents' happiness. When we added marital status to the benchmark in column 5, there was no significant change in the coefficient as it decreased by only 0.0007. This shows that education has less effect on rural residents' happiness through marriage status, perhaps because the spouses of rural residents are acquaintances in the same village or are colleagues at work. Their education levels are largely similar; therefore, there is no need to screen respondents based on their education levels in the marriage scenario. Moreover, only 1.94 % of respondents had a college education or higher, which makes the effect of education on finding a suitable spouse in the marriage market relative less significant. In summary, except for marriage pathways, empirical results support [Hypothesis 5](#) by showing that rural residents' happiness is partly influenced by education but through income and health

Table 6

Annual PSM-DID regression results by different income groups.

	(1)	(2)	(3)	(4)	(5)
	Low income	Middle income	High income	Extremely low income	Lower income
Policy	−0.5767** (−2.1512)	−0.3469** (−2.2804)	0.1824*** (3.7141)	−0.8726*** (−6.6369)	−0.0037 (−0.0158)
Control variables	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes
Regional fixed effects	Yes	Yes	Yes	Yes	Yes
Urban clustering	Yes	Yes	Yes	Yes	Yes
Observations	6238	5509	5599	3801	3397
Adj. R ²	0.0750	0.0755	0.0591	0.0916	0.0710

In columns 1 through 3 of this table, the annual household income is divided into 33.3 % and 66.7 %, representing low-income, middle-income and high-income families, respectively. In columns 4 through 5 of this table, the annual household income is divided into five points, representing extremely low income and lower income. T-statistics are reported in parentheses. *, **, and *** indicate significance at the 10 %, 5 %, and 1 % level, respectively.

Table 7

Annual PSM-DID regression by different education level groups.

	(1)	(2)	(3)	(4)	(5)
	No education	Primary education	Junior high school education	High school education	College education and above
Policy	−0.3952*** (−3.7876)	−0.9006*** (−3.8851)	−0.5866*** (−3.3556)	0.7528*** (3.0425)	−0.1175 (−0.3501)
Control variables	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes
Regional fixed effects	Yes	Yes	Yes	Yes	Yes
Urban clustering	Yes	Yes	Yes	Yes	Yes
Observations	3581	5977	5963	1568	257
Adj. R ²	0.0723	0.0809	0.0733	0.0927	0.1448

T-statistics are reported in parentheses. *, **, and *** indicate significance at the 10 %, 5 %, and 1 % level, respectively.

Table 8

Effects of education on employment, income, health, and marriage.

	(1)	(2)	(3)	(4)
	Employment status	Personal income	Health status	Marital status
EdU level	0.0830*** (18.8142)	0.0365*** (13.2568)	0.0139*** (5.0475)	0.0027*** (3.2332)
Other variables	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Regional fixed effects	Yes	Yes	Yes	Yes
Urban clustering	Yes	Yes	Yes	Yes
Observations	15,586	15,586	15,586	15,586
Adj. R ²	0.2885	0.0339	0.2452	0.3601

T-statistics are reported in parentheses. *, **, and *** indicate significance at the 10 %, 5 %, and 1 % level, respectively.

channels rather than marriage channels.

4.4. Robust tests

4.4.1. Parallel trend test

One prerequisite for the validity of the DID estimation is passing the parallel trend test. To ensure that the difference in happiness between the experimental and control groups is attributable to the 12-year free-education policy, we follow Beck et al. (2010) and used the event study method to measure the dynamic effect of the 12-year policy implementation. To mitigate bias from collinearity, we eliminated the base period. As shown in Fig. 2, before the policy was implemented, the coefficient was insignificant (i.e., the 95 % confidence interval crosses a horizontal dotted line where the coefficient equals 0), while it differs significantly from 0 in the year of policy implementation and later years. This shows that the experimental and control groups before policy implementation did not significantly differ in terms of wellbeing, thereby passing the parallel trend test.

4.4.2. Placebo test

To assess the potential impact of nonobservable factors on the happiness effect of the 12-year free-education policy, we used a placebo test method based on the method by La Ferrara et al. (2012). This involved generating 500 random city-level policy impacts using computer simulation. As the “pseudo-treatment” group was randomly assigned, any resulting policy shock would not have an actual impact on happiness, that is, $\beta^{\text{random}} = 0$. Therefore, in the absence of significant omitted variable bias, the regression coefficient should be approximately zero. Conversely, a statistically significant deviation from zero would indicate an identification bias in our model. Fig. 3 presents the distribution of estimated coefficients and corresponding p-values for these 500 randomly generated treatment groups. The scatter points for p-values are mainly clustered near 0 (0.00008) and far from their true value (−0.415). Furthermore, most scatter points fall above the horizontal dotted line at a p-value of 0.1, indicating insignificance at the 10 % level. Thus, it can be concluded that other unobserved factors do not significantly affect the happiness seen among rural residents participating in the 12-year free-education policy pilot program.

4.4.3. Covariate balance test

When selecting covariates for PSM, we must conduct a balance test on the covariates of the experimental and control groups to ensure the randomness and exogeneity of the policy reform and enhance the validity of the regression results. Thus, we conducted a period-by-period balance test on the effect of PSM and showed that covariate coefficient values in each year after matching declined significantly and gradually became insignificant. The R² value also decreased for all cases, indicating effective identification matching through the covariate balance test.

5. Conclusions and policy recommendations

Free education is a crucial mechanism for promoting equal educational opportunities and realizes the power of education in China.

Table 9
Effects of employment, income, health and marital status on rural residents' happiness.

	(1)	(2)	(3)	(4)	(5)
	Baseline regression	Personal income (no employment)	Personal income (plus, employment)	Health status	Marital status
EdU	0.0268*** (8.6269)	0.0203*** (6.1461)	0.0018*** (5.3586)	0.0216*** (6.6499)	0.0261*** (8.2906)
lnIncome		0.1653*** (13.1953)	0.1622*** (12.9935)		
Employ			0.0235*** (2.6254)		
Health				0.2624*** (23.3514)	
Marriage					0.2586*** (6.8620)
Other variables	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes
Regional fixed effects	Yes	Yes	Yes	Yes	Yes
Urban clustering	Yes	Yes	Yes	Yes	Yes
Observations	17,364	15,594	15,593	17,355	17,364
Adj. R ²	0.0295	0.0379	0.0381	0.0543	0.0311

T-statistics are reported in parentheses. *, **, and *** indicate significance at the 10 %, 5 %, and 1 % level, respectively.

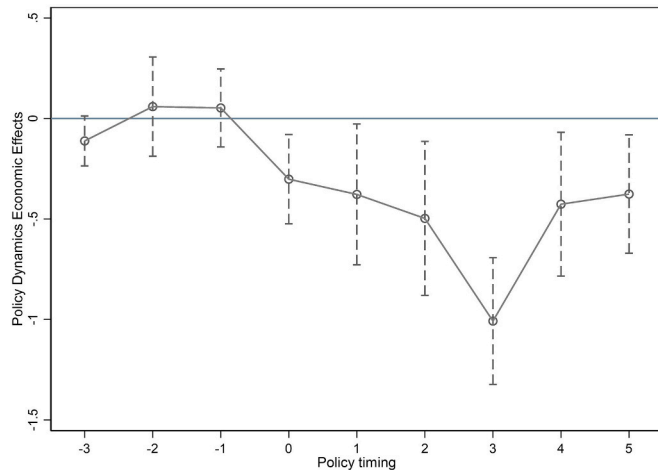


Fig. 2. Parallel trend test.

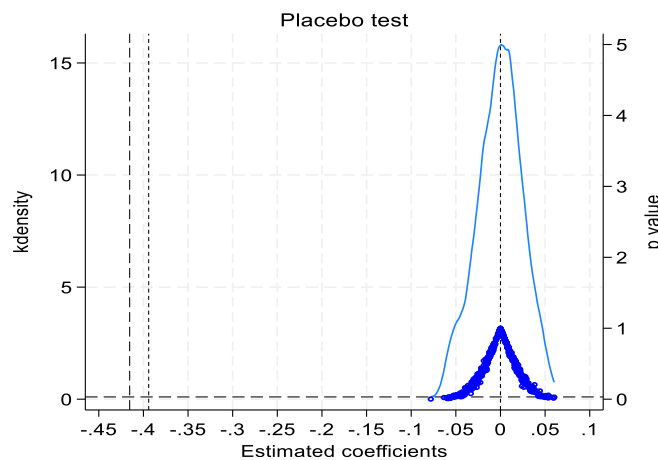


Fig. 3. Placebo test.

Currently, the debate over whether the period of compulsory education should be extended from 9 to 12 years is a matter of public and the government concern. Using cross-sectional data from the fifth phase of the CGSS from 2010 to 2015, we used the PSM-DID method to examine the impact of a 12-year free-education policy pilot on rural residents'

happiness for the first time to provide empirical evidence for the choice of the duration of 12 years of compulsory education from the happiness perspective. The empirical results support the findings of [Zhu and Yu \(2023\)](#) and [Jin and Liu \(2023\)](#) and show that the government's inadequate financial resources and limited policy benefits of the 12-year free-education policy has reduced the rural residents' happiness. We examined the regional effect of the pilot policy and found that its implementation in Zhangzhou and Tangshan City improved the happiness of rural residents mainly because these governments had sufficient financial resources, which allowed them to sustain other public services despite the policy implementation. The heterogeneity analysis results suggest that the policy mainly improved the rural beneficiaries' happiness, including grandparents, higher-income groups, and higher educated groups. The mechanism test indicates that education influences rural residents' happiness through income and health paths but weakly through marriage paths. We also conducted parallel trend, placebo, and covariate balance tests to confirm the robustness of our results.

As the economy and society continue to develop, people's educational needs will grow gradually. According to the sufficiency theory of educational justice, the government must extend the duration of compulsory education. However, this study shows that only when the number of beneficiaries of the 12-year free-education policy increases and when central and local governments can support the reform, the 12-year compulsory education reform may be successfully implemented in China. Therefore, the scope of the policy should be further extended. We make the following suggestions.

First, the mechanism for providing free high school education subsidies to impoverished rural families needs improvement. The study findings suggest that low-income rural families face significant barriers to benefiting from the policy because of the lack of awareness about education and the inability to afford the added costs of high school education. According to the educational justice theory of prioritization, disadvantaged children from lower-income families should receive the resources. The subsidy standards for high school education should be aligned with the "two exemptions and one subsidy" policy for compulsory education. Eligible children from rural families should receive subsidies for boarding living expenses, meals, and auxiliary teaching materials to reduce their families' financial burden and thereby reduce their sense of relative deprivation and improve their family's overall wellbeing.

Second, stable financial support is a prerequisite for implementing the 12-year free-education policy. It is crucial to improve the local tax system and expand local taxation management authority appropriately and actively explore stable local main tax types such as property tax to

provide a steady source of revenue for local governments. Alternatively, the central government can offer financial aid for implementing and extending the policy to rural areas through transfer payments and special funds. Educational egalitarianism asserts that rural children should have equal educational opportunities as urban children. Therefore, to achieve true equality of opportunity, resources should be directed toward rural areas and subsidies must be offered to provinces with a large population base and financial difficulties based on the per capita standard. Moreover, the central government should allocate pilot funds to some extremely poor counties and cities to achieve a balanced development of high school education across different regions.

Third, it is essential to establish welfare-oriented education decision making and adjust fund allocation dynamically for various public services. The government should clarify its welfare goals in the field of educational public services and establish suitable measurement indicators. However, owing to differences in realizing welfare goals through different education policies, determining which education fields require further development is imperative. This will allow for dynamic adjustment of public service funding allocation priorities, ensuring investment in key areas. Different public services have diverse effects on rural residents' happiness (Table 9); education affects rural residents' in different ways, such as employment, income, and health. Therefore, it is important to coordinate various public services to achieve comprehensive resource use. For example, when increasing investment in education, it is beneficial to coordinate education with medical services through health education and campus medical treatment to improve the efficient use of financial resources and maximize rural residents' happiness.

Due to the limitations of local and municipal CGSS data, this study used data from 2010 to 2015, which limits the policy beneficiaries in the short term and obscures the positive effect of the policy. Moreover, due to the lack of proper longitudinal survey data, we focused on the indirect and potential beneficiaries of the 12-year free-education policy, resulting in a lack of measurement of the happiness effect of students beneficiaries. Future research on the happiness effects of 12-year compulsory education policy could benefit from the availability of long-term data, data on broader populations, and longitudinal tracking surveys in terms of scope and accuracy.

Data availability

Data will be made available on request.

Acknowledgments

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